

Differential Microscopic Mobility of Components within a Deep Eutectic Solvent

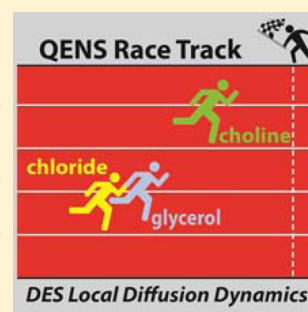
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S Supporting Information

ABSTRACT: From macroscopic measurements of deep eutectic solvents such as glyceline (1:2 molar ratio of choline chloride to glycerol), the long-range translational diffusion of the larger cation (choline) is known to be slower compared to that of the smaller hydrogen bond donor (glycerol). However, when the diffusion dynamics are analyzed on the subnanometer length scale, we find that the displacements associated with the localized diffusive motions are actually larger for choline. This counterintuitive diffusive behavior can be understood as follows. The localized diffusive motions confined in the transient cage of neighbor particles, which precede the cage-breaking long-range diffusion jumps, are more spatially constrained for glycerol than for choline because of the stronger hydrogen bonds the former makes with chloride anions. The implications of such differential localized mobility of the constituents should be especially important for applications where deep eutectic solvents are confined on the nanometer length scale and their long-range translational diffusion is strongly inhibited (e.g., within microporous media).



Deep eutectic solvents (DESs)^{1,2} have emerged as low-cost, not-toxic, biodegradable alternatives to the more traditional room temperature ionic liquids (RTILs). They are promising candidates for a broad range of applications including chemistry, engineering, nanoscience, and biomass processing.³ In contrast to traditional RTILs, which exclusively contain ions, the constituents of a DES include a *molecular* hydrogen bond donor species, in addition to ions. Development of a hydrogen bonding network within a DES is essential for eutectic formation and can be expected to profoundly influence the structural and dynamical aspects of a DES. Understanding the microscopic diffusion dynamics, which ultimately define the macroscopic transport properties, is central to the intelligent application of DESs as sustainable solvents. Despite this fact, investigations of microscopic dynamics within DESs have received surprisingly little attention to date. Pulsed field gradient NMR studies of an archetypical DES glyceline⁴ (1:2 molar ratio of choline chloride to glycerol) and other common DESs⁵ have revealed faster long-range diffusion for neutral hydrogen bond donors, such as glycerol, compared to that for the choline cation. This has been rationalized on the basis of the relatively larger size of the latter species. While pulsed field gradient NMR is a valuable tool for probing long-range mobility of the constituents,⁶ for applications wherein DESs are strongly confined, with impeded long-range translational mobility, the most relevant microscopic diffusion dynamics should be probed directly on the nanometer and subnanometer length scales. Quasielastic neutron scattering (QENS) is notably up to this task because the neutron wavelength (and thus the scattering momentum transfer) is

comparable with the interparticle distances, while the QENS energy transfer range corresponds to the relevant pico- to nanosecond time scale. Moreover, QENS allows us to harness another remarkable property of neutrons as a scattering probe: namely, the large incoherent scattering cross-section of hydrogen compared to all other elements, including deuterium. On the basis of the neutron scattering cross sections, for the mixture of choline chloride with perdeuterated glycerol used in the current experiment, the relative contribution of choline chloride to the total incoherent scattering signal was 97%. On the other hand, for the mixture of deuterated choline chloride with glycerol used in the current experiment, the relative contribution of glycerol to the total incoherent scattering signal was 92%. Therefore, when studying otherwise similar glyceline mixtures, one containing deuterated choline and another with perdeuterated glycerol, we essentially probe the dynamics of glycerol in the former sample and the dynamics of choline in the latter. That is, selective deuteration (an “isotope-edited” approach) allows us to interrogate the spatial dependence of the diffusion dynamics separately for the cation and the hydrogen bond donor species within a DES. The schematic structures of the selectively deuterated DESs are presented in Figure S1 of the Supporting Information along with their characterization results by NMR.

Figure 1 shows the temperature dependence of the scattering intensity (dominated by protons of either hydrogenated choline

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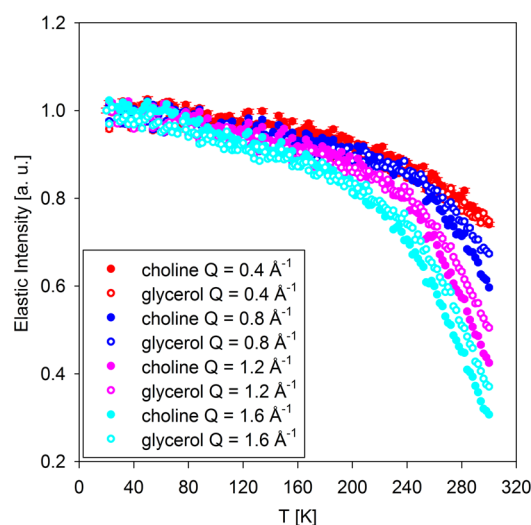


Figure 1. Elastic incoherent scattering intensity from the glyceline components measured at various scattering momentum transfer values, Q , as a function of temperature upon heating. The raw data were normalized to unity at 20 K. When not visible, the error bars are within the symbols.

or hydrogenated glycerol and thus predominantly incoherent) that appears elastic within the experiment resolution window of $3.4 \mu\text{eV}$, which corresponds to ca. 0.4 ns . In such a measurement, the incoherent scattering intensity will appear elastic if, within the time corresponding to the experiment energy resolution, $\tau \sim 1/E$, a moving particle remains within a volume with dimensions related to the scattering momentum transfer, $d \sim 1/Q$. One can immediately make several observations from the results summarized in Figure 1, as follows.

- (1) Glyceline is a glass-forming liquid that does not exhibit visible signs of crystallization upon heating up from the glassy state at the baseline temperature. Traditional RTILs sometimes fail to crystallize on cooling down, but typically exhibit crystallization when heated from the glassy state.^{7–10} The stronger glass-forming propensity of DESs compared to traditional RTILs has been noted in the literature.¹¹ Additional characterization of the physical state of the samples is included in the Supporting Information (Figure S2, S3, S4).
- (2) The microscopic dynamics of glyceline's constituents exhibit only vibrational degrees of freedom in the glassy state. Above the glass transition temperature, the microscopic dynamics are dominated by diffusive motions yielding a strongly Q -dependent signal, as could be expected for a liquid.
- (3) Most importantly in the context of the present study, choline cations exhibit consistently less elastic scattering than glycerol molecules. Such decreased elastic scattering is directly related to the enhanced local displacements. Therefore, the data in Figure 1 represent a direct, completely model-free proof of the enhanced local diffusive dynamics of choline compared to glycerol within the DES glyceline.

To extract quantitative information on the comparative diffusive dynamics of choline and glycerol, we turn to the QENS spectra. In contrast to previous QENS studies of RTILs,^{10,12–14} we could not satisfactorily fit the data using two

Lorentzian terms and background (see Figure S5 of the Supporting Information). Instead, three Lorentzian terms were required. The necessity of the third Lorentzian term, even at a low Q , where the relaxation processes are usually Debye-like and not stretched, is rather unusual. While it may merely be a manifestation of the fast internal reorientation dynamics of the side groups, this might alternatively be attributed to the complex nature of the three-component glyceline. It is possible that the chloride anion, which participates in hydrogen bonding, attracts sufficient proton density from both choline and glycerol, thereby yielding an appreciable signal associated with its relatively fast diffusional dynamics regardless of which glyceline component is deuterated. In any case, the fastest of the three fit components, while necessary for adequate fitting, was too broad for the spectrometer dynamic range and could not be reliably analyzed. The remaining two components, however, can be reliably analyzed in detail using the approach customarily applied to two-component data fitting for QENS data obtained from liquids.

The necessity of the two-component description of the QENS data collected from liquids, both complex and simple, along with its origin and implications, has been extensively described in the literature.^{10,15–17} Since these two distinct components, which we denote fast (localized) and slow (long-range), are always present within liquid QENS spectra,^{10–17} their origin should not be attributed to different species in the sample in the current experiment. Instead, both components originate from the center-of-mass diffusive dynamics of either choline (in the case of glyceline containing perdeuterated glycerol) or glycerol (for glyceline containing deuterated choline). The faster of the two components is ascribed to the localized diffusive motions confined in the transient cage of neighbor particles, which precede the cage-breaking long-range diffusion jumps associated with the slower of the two diffusion components. The relative spectral weight of the slower component then represents the elastic incoherent structure factor of the localized in-cage motions, thus potentially yielding the characteristic localization radius of such motions.

The Q -dependence of the fit components affirms the validity of this approach. Analogously to previous QENS studies of RTILs,^{12–14} the QENS data could be fitted up to the values slightly above $Q = 1 \text{ \AA}^{-1}$, that is, below the structural factor maximum. The width of the narrower component approaches zero in the limit of low Q , as shown in the bottom inset of Figure 2, thus representing long-range translational diffusion. At each given temperature, we estimated the translational diffusion coefficient, D , from the low- Q slope of the Q -dependence of the half-width of the Lorentzian half-maximum using the relationship $\text{HWHM}(Q) = \hbar D Q^2$, where $\hbar$ is the reduced Planck constant. On the other hand, the width of the broader component plateaus in the limit of low Q , as shown in the top inset of Figure 2, thus representing localized in-cage motions inside the transient cage of the nearest neighbors. For such dynamics, the localized diffusion coefficient can be evaluated¹⁸ from the low- Q plateau value of the half-width of the Lorentzian half-maximum as $\text{HWHM}(Q) = 4.33\hbar D/a^2$, where a is the characteristic confinement radius (although, in the case of the in-cage dynamics in liquids, the confinement is transient). The transient confinement radius can be evaluated most accurately from fitting the relative spectral weight of the narrow Lorentzian, $p(Q)$, as shown in Figure 3 and its inset. The $p(Q)$ can be fitted as elastic incoherent structure factor for the motions in the in-cage transient confinement using a

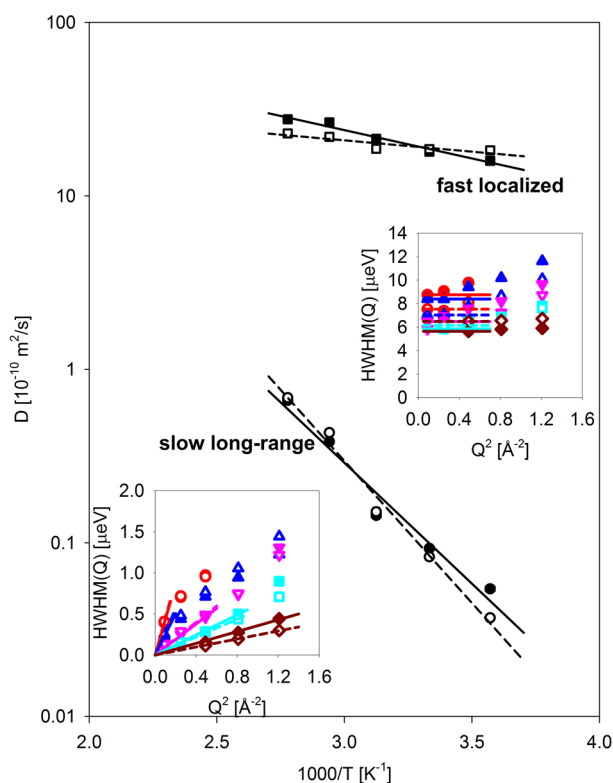


Figure 2. Temperature dependence of the diffusion coefficients extracted from the Q -dependence of the QENS fit component widths using an expression $\text{HWHM}(Q) = \hbar D Q^2$ for the narrow and $\text{HWHM}(Q) = 4.33 \hbar D / a^2$ for the broad component, as shown in the insets. Filled symbols, solid-line Arrhenius fits: choline in glyceline. Open symbols, dashed-line Arrhenius fits: glycerol in glyceline. Red: 360 K. Blue: 340 K. Pink: 320 K. Cyan: 300 K. Brown: 280 K. The error bars are within the symbols.

functional form for diffusion inside a sphere,¹⁸ $[3j_1(Qa)/(Qa)]^2$, where j_1 is the spherical Bessel function of the first order, and a is the transient confinement radius of our particular interest.

As one can see in the inset of Figure 3, there is a substantial difference in the $p(Q)$ between choline and glycerol in glyceline at 280 K, which progressively decreases and practically disappears at the higher temperature. Reflecting this temperature dependence, the transient confinement radii for choline and glycerol show divergence as the temperature is decreased, besides becoming progressively smaller. In agreement with the elastic data from Figure 1, choline experiences roomier local transient confinement compared to glycerol.

The long-range diffusion in DESs has been traditionally analyzed using Hole theory,^{4,5,19–22} which considers a comparison between the size of a void available for a diffusional jump and the hydrodynamic radius of the diffusing species. The latter is typically evaluated from macroscopic diffusion data using a Stokes–Einstein expression. The long-range diffusion coefficients and their activation energies for choline and glycerol from the data presented in Figure 2 are comparable with the glyceline data reported in the literature.^{4,5} In particular, we obtain activation energies of (26.7 ± 2.9) kJ/mol and (31.3 ± 2.0) kJ/mol for the long-range diffusion of choline and glycerol, respectively, to be compared with the value of ca. 42 kJ/mol from the NMR measurements.⁵ However, the faster long-range diffusion of glycerol compared to choline, as

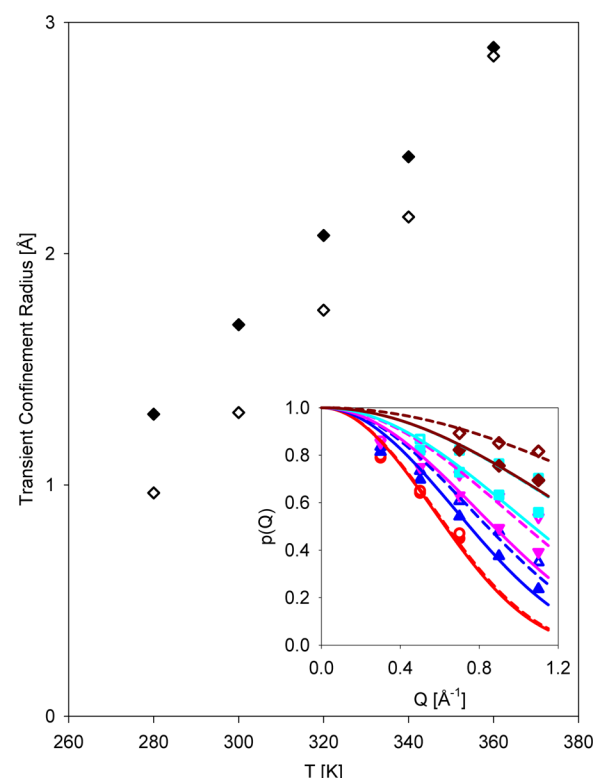


Figure 3. Temperature dependence of the transient confinement radius extracted from the Q -dependence of the spectral weight of the narrow QENS fit component using an expression $p(Q) = [3j_1(Qa)/(Qa)]^2$, as shown in the inset. Filled symbols, solid lines: choline in glyceline. Open symbols, dashed lines: glycerol in glyceline. Red: 360 K. Blue: 340 K. Pink: 320 K. Cyan: 300 K. Brown: 280 K. The error bars are within the symbols.

reported in the literature,^{4,5} suggests the larger hydrodynamic radius of the latter,^{4,5} in agreement with the relationship between the hard sphere radius of 3.29 Å for choline and 3.00 Å for glycerol. If choline is consistently larger in size than glycerol, how is it possible for choline to be less spatially confined, as evident from Figure 3, and concurrently exhibit larger localized displacements, as illustrated by the model-independent data presented earlier in Figure 1?

To address this question, we turn to simulation. The molecular orientations and the interactions between choline, chloride ion, and glycerol molecules are crucial for understanding the mobility of species within glyceline. Our simulations reveal that the chloride anion plays a pivotal role in the formation of the hydrogen bonded network, which is central to the “deep eutectic” nature of the glyceline DES. A competitive hydrogen bonding activity is observed between glycerol and choline cation for binding with chloride anion. It turns out that glycerol forms tighter hydrogen bonds with the chloride anion via multiple hydroxyl groups to form a closed-ring type structure (i.e., hydrogen bonding between a vicinal diol unit of glycerol and chloride is evident in Figure 4), outcompeting the lone hydroxyl group presented by the choline moiety. The energy-minimized structure shown in Figure 4 is an indication of stronger interactions present between glycerol and the chloride anion, compared with choline–chloride interactions. Moreover, relative to the calculated geometry for pure choline chloride, the distance between the chloride anion and the choline headgroup increases to a large extent for

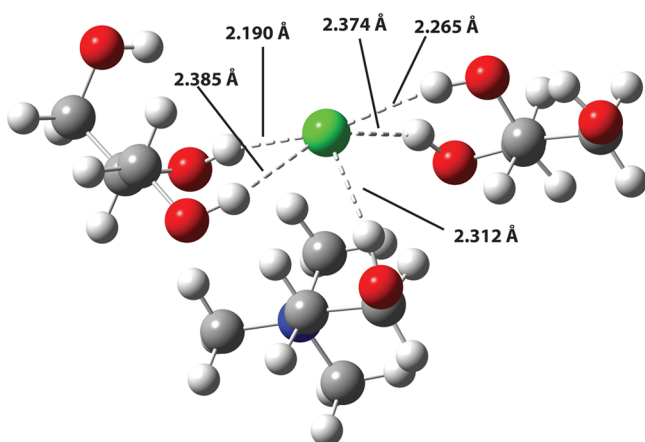


Figure 4. Simulation results revealing that the hydrogen bonds formed between glycerol molecules and chloride dominate over those established between the choline cation and chloride. Atoms are colored according to type: C (gray), H (white), O (red), N (blue), and Cl (green).

glyceline, indicative of reduced cation–anion interaction within the eutectic mixture. Our simulation results thus provide an explanation for the more restricted local diffusive mobility of glycerol, despite its smaller size compared to choline. Compared to the choline cation, glycerol is much more involved with or “tied up in” the formation of hydrogen bonding networks in the DES glyceline.

In summary, our QENS experiments have revealed that choline cations in the DES glyceline experience looser transient confinement than the hydrogen bond donor glycerol, despite the larger size of choline compared to glycerol. Our simulation results allow us to explain this observation by demonstrating that glycerol outcompetes choline for hydrogen bonding with chloride anion. This leads to a differential localized mobility of the DES constituents, wherein choline experiences less restrictive local transient confinement than glycerol. While long-range diffusivity is clearly important for mass transport in bulk DESs, the localized mobility should play a prominent role in the chemical and biochemical applications of DESs involving confinement on the nanometer length scale. In such environments, the long-range translational diffusion is strongly suppressed, and the diffusive dynamics are dominated by the localized motions that are not strongly affected by the confinement of the solvent. As a final point, this investigation demonstrates the power of quasielastic neutron scattering to elucidate short-range diffusional dynamics within complex solvents, especially when paired with selective deuteration (isotopic editing) to provide signal discrimination for the desired solvent component.

METHODS

Chemicals. The deuterated precursors choline- d_{13} chloride and glycerol- d_8 used to prepare the selectively deuterated DESs needed to achieve neutron contrast in our experiments were acquired from C/D/N Isotopes Inc. (Quebec, Canada). The protiated versions of choline chloride and glycerol were sourced from Sigma-Aldrich (St. Louis, MO).

Preparation of Deuterated DESs. Partially deuterated (i.e., isotope-edited) DESs (denoted D1 and D2) were prepared by mixing a 1:2 molar ratio of choline chloride to glycerol in which one of the eutectic components was deuterated and the other

remained hydrogenated. The D1 systems comprises deuterated choline and protiated glycerol, allowing us to selectively probe the dynamics of glycerol, whereas the eutectic mixture D2 contains perdeuterated glycerol mixed with choline- d_1 chloride (which encompasses 13 protons), making the isolation of choline dynamics possible.

Preparation of Choline- d_1 Chloride. Choline- d_1 chloride was prepared by performing deuterium exchange of the hydroxyl proton on choline by dissolving choline chloride (0.5 g) in 5 mL of deuterated methanol (CD_3OD , 99.96 atom % D), followed by solvent removal under vacuum. This process was repeated three times to ensure complete deuterium exchange and was verified using FTIR and NMR ($DMSO-d_6$). 1H NMR (300 MHz, D_2O): δ 4.18 (2H, m), 3.57 (2H, dd), 3.25 (9H, s).

Preparation of D1 (1:2 Choline- d_{13} Chloride:Glycerol). D1 was prepared by mixing choline- d_{13} chloride (0.360 g) and glycerol (0.241 g) in a 1:2 mol ratio. The mixture was then stirred, gradually heating to 80 °C, followed by drying under vacuum on a Schlenk line at 50 °C overnight to remove residual moisture. 1H NMR (300 MHz, D_2O): δ 3.82 (1H, m), 3.65 (4H, ddd). 1H NMR spectra for D1 and D2 are presented alongside that for the fully protiated glyceline in Figure S1 of the Supporting Information.

Preparation of D2 (1:2 Choline- d_1 Chloride:Glycerol- d_8). D2 was prepared in the same manner as for D1 using choline- d_1 chloride (0.380 g) and glycerol- d_8 (0.285 g). 1H NMR (300 MHz, D_2O): δ 4.18 (2H, m), 3.57 (2H, dd), 3.25 (9H, s).

Quantum Chemical Calculations. The optimized geometry of the DES mixture was obtained using various starting geometries wherein two glycerol molecules were placed around choline chloride at different orientations. The initial geometry optimizations were done at HF/6-311G(d,p) followed by geometry optimization of the DES mixture at the M062x level of theory using Pople’s medium 6-31++g(d,p) basis set. The optimized structures were tested for the stability of their wave function and any negative eigenvalues resulting from transition state structures or higher order saddle points, using stable and opt=calcf procedures, respectively. The absence of imaginary frequencies in the calculated vibrational frequency of the optimized structures ensured stable structure. The calculated interaction energy at M062X/6-31++g(d,p) was within 2 kcal/mol of the energy calculated using MP2/aug-cc-PVDZ theory which indicates reliability of the data. All the calculations reported here were performed using the Gaussian 09 program package.²³

QENS Measurements. The liquid samples were loaded in annular aluminum sample holders with an outer diameter of 29 mm and an inner diameter of 28.8 mm. The resulting gap of 0.1 mm was chosen to limit the effects of multiple scattering. The sample holders were sealed with indium gaskets and placed in a close-cycle refrigerator providing a temperature stability of better than 0.5 K. The neutron measurements were performed at the SNS backscattering spectrometer, BASIS.²⁴ The energy resolution, measured individually for each sample at a baseline temperature of 7 K, showed a Q-averaged value of 3.4 μ eV (full width at half-maximum), while the dynamic range of energy transfer of ± 100 μ eV was used for the data analysis. The quick elastic intensity scans were collected on warming up from the baseline temperature following the resolution measurements. The dynamic data were then collected at 360, 340, 320, 300, and 280 K. The final spectra obtained using standard MANTID-based routines were analyzed using the DAVE software package.²⁵

■ ASSOCIATED CONTENT

■ Supporting Information

Methodology details, results of NMR characterization and physical state characterization, technical discussion of the QENS data fitting, and supplementary figures are supplied. The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpclett.5b01192.

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Notes

The authors declare no competing financial interest.

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